

USB-CAN Bus Adapter (CANalyst-II Analyzer)

Out-of-The-Box Test Instructions

Instruction Manual Version: V2.04

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I. Introduction

First of all, thank you for purchasing and using our products, our company will always be happy to serve you!

This document is to guide the first time buyers to test whether the USB-CAN bus adapter/CANalyst-II analyzer can work properly.

It can be ignored by experienced users or users who are more familiar with the operation of the USB-CAN Bus Adapter/CANalyst-II.

II. Test Steps

When you receive the USB-CAN bus adapter/CANalyst-II analyzer, please follow the steps described below in order to test it.

1. Install the Driver for the Hardware

Connect the USB-CAN bus adapter/CANalyst-II analyzer to any USB port of your computer with the standard accessory "USB cable", and the computer will pop up the "Driver Installation Wizard".

Please refer to *"2.USB Driver Installation and Uninstallation Manual.pdf"* for detailed steps on driver installation.

2. Installation of Test Software

For detailed step-by-step instructions on the installation of the test software, please refer to *"3. USB-CAN Tool Debugging Software Installation and Usage Manual.pdf"*.

3. Connecting the Two Channels of the USB-CAN Bus Adapter

3.1 Shorting the Termination Resistor

Short the R+ and R- of the CAN1 channel of the USBCAN adapter with a wire and the R+ and R- of the CAN2 channel with a wire.

Toggle R1 and R2 in the blue dip switches of the CAN1 channel on the left side of the CANalyst-II analyzer to the upper OFF position; and Toggle R1 and R2 in the red dip switches of the CAN2 channel on the right side of the CANalyst-II analyzer to the lower ON position.

Note: Be sure to configure the termination resistor before connecting the device to the USB port of the computer via the USB cable, otherwise the test will not pass,

and you need to plug and unplug the USB to retest.

3.2 Connect the CANH and CANL Signal Lines of CAN1 and CAN2 Channels.

Short the CANH of CAN1 channel and CANH of CAN2 channel, and short the CANL of CAN1 channel and CANL of CAN2 channel. The schematic diagram after shorting is shown below:

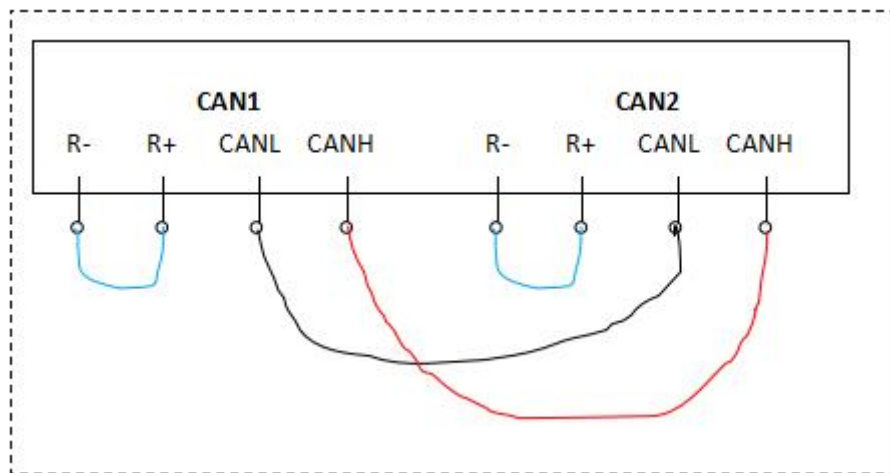


Figure 1 USBCAN Bus Adapter Test Wiring Diagram

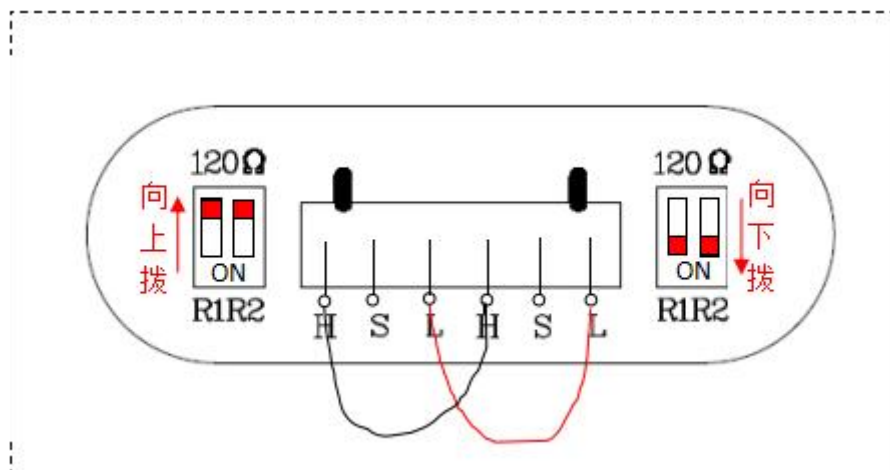


Figure 2 CANalsyt-II Analyzer Test Wiring Diagram

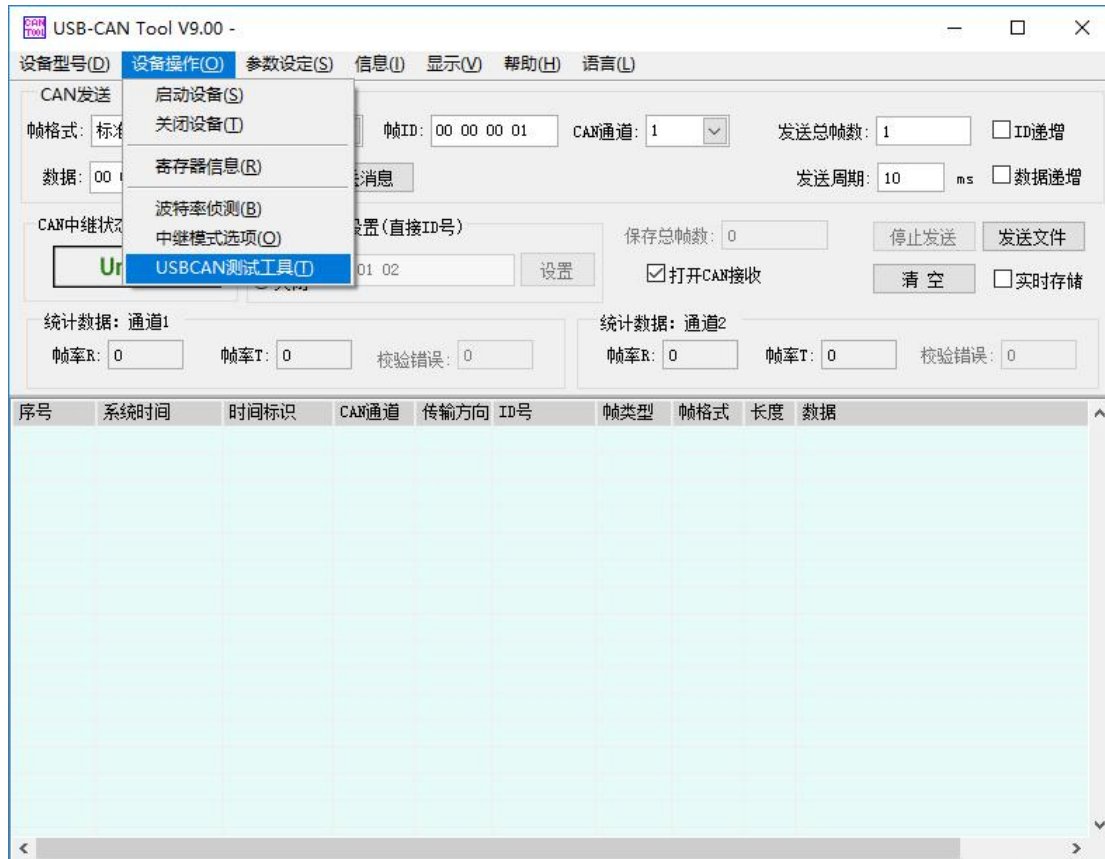
4. Run the USB-CAN Tool Test Tool

Run USB-CAN Tool debugging software, if the computer is installed in the protection software, such as 360 security guards, may open the exe program, pop-up dialog box prompts the program to modify the registry entries, allow can be, which is LabVIEW running engine in the running of some of the program's operations:

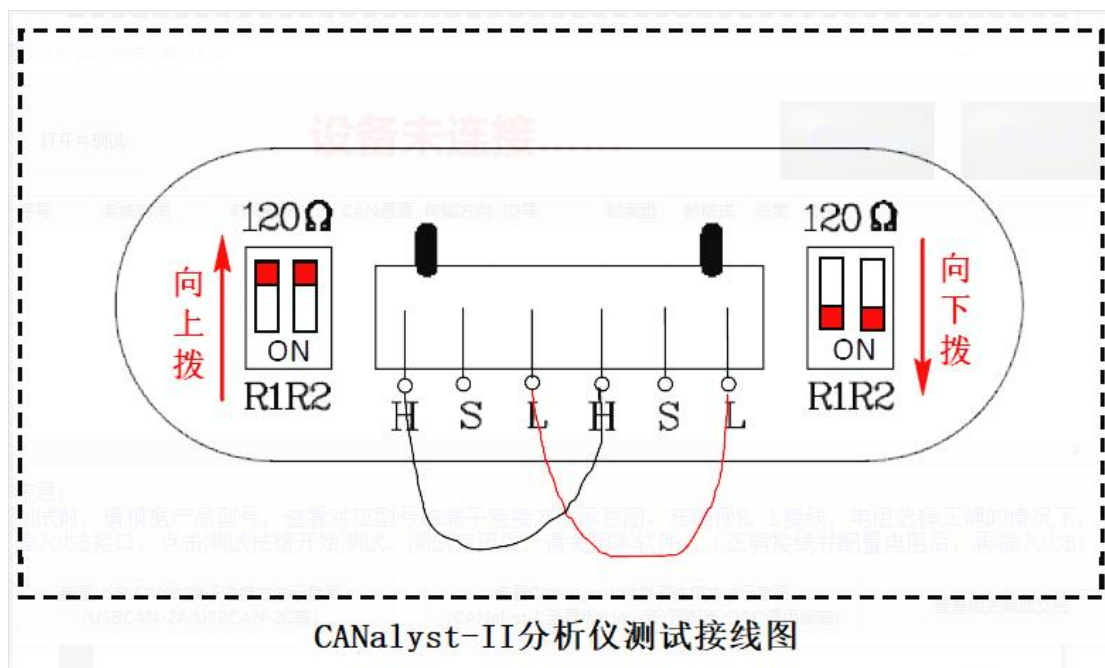
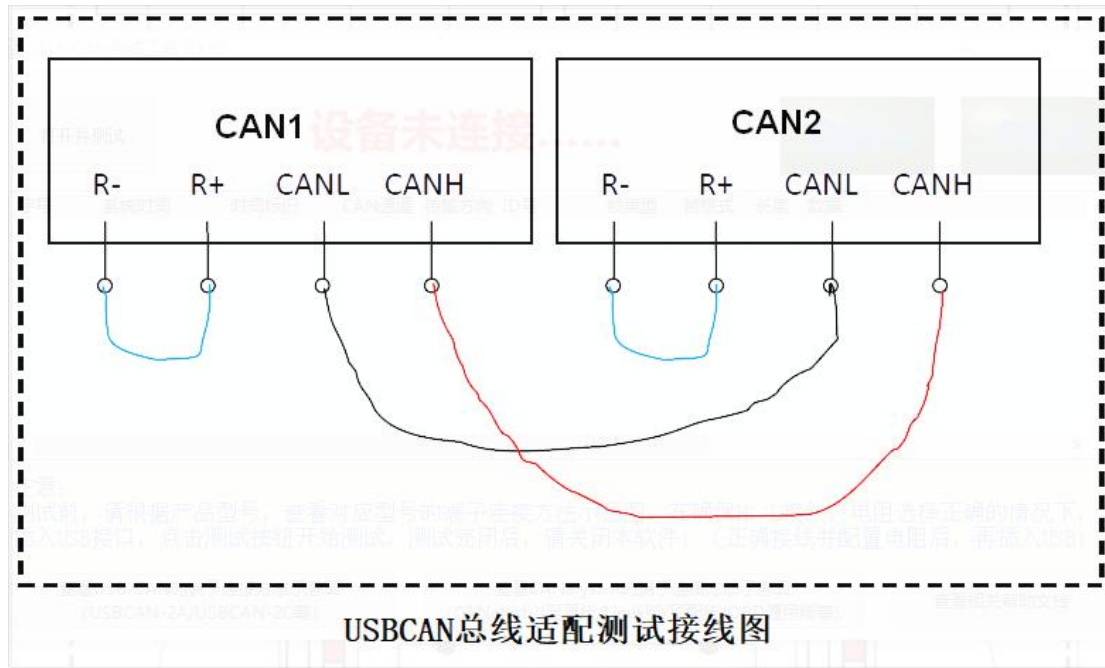


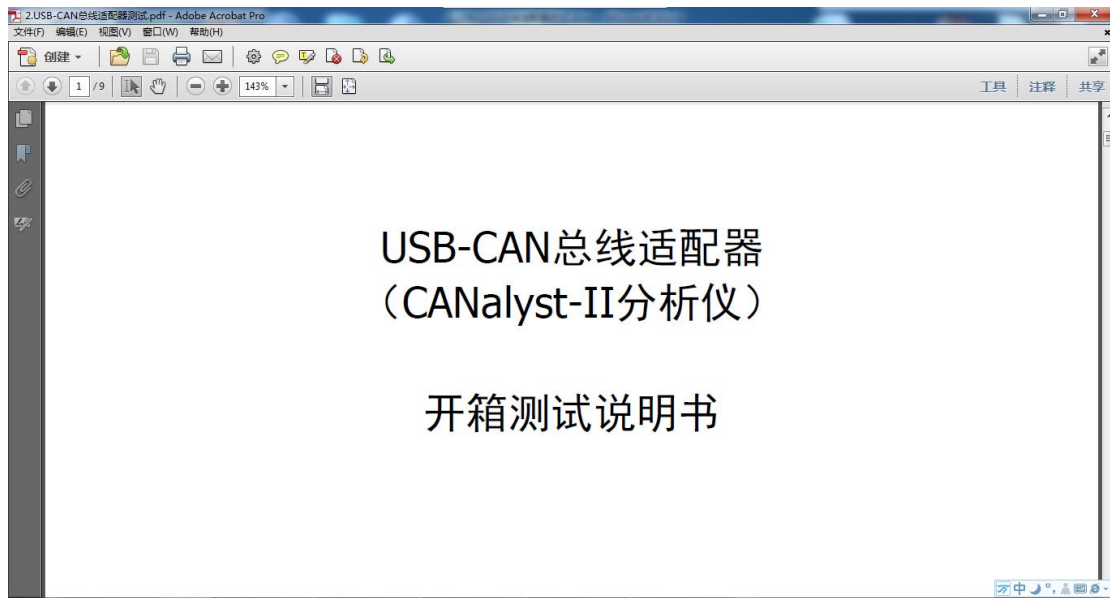
4.1 Automatic Testing With the USBCAN Test Tool

In the menu of the USB-CAN Tool interface, click on "Device Operation USBCAN Test Tool":



Click on the three buttons below to view wiring diagrams, manuals for the corresponding model.





Connect the wires according to the diagram, plug in the device, click on the upper left corner of the "open and test" button, the software will automatically test (in order to open the device -> initialize CAN1/CAN2 two channels -> CAN1 sends a sequence, CAN2 to receive and verify -> CAN2 sends a sequence, CAN1 to receive and verify -> turn off the device -> the results are displayed):



If the following result appears, the test passes:

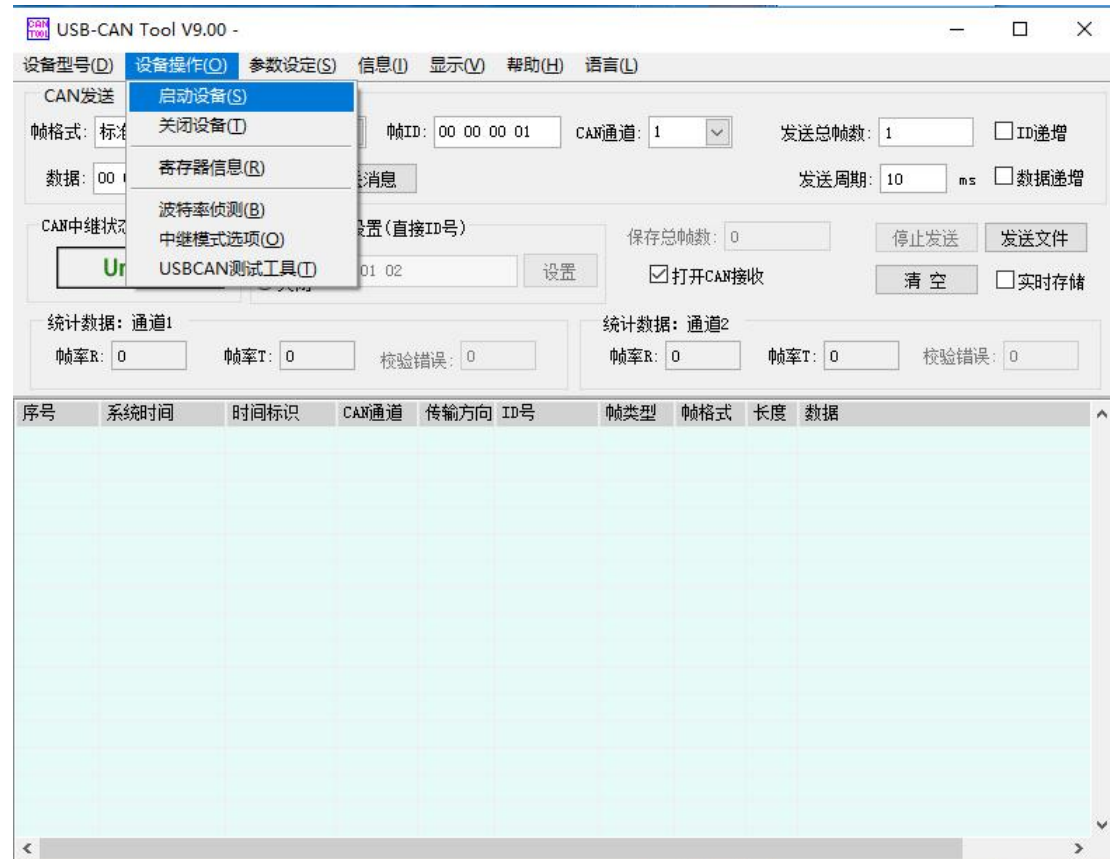


If the following results occur, follow the prompts to check wiring, terminating resistor configuration, and plug and unplug the USB to retest:



4.2 Manual Test

In the pop-up menu of USB-CAN Tool interface, click "Device Operation" to start the device:



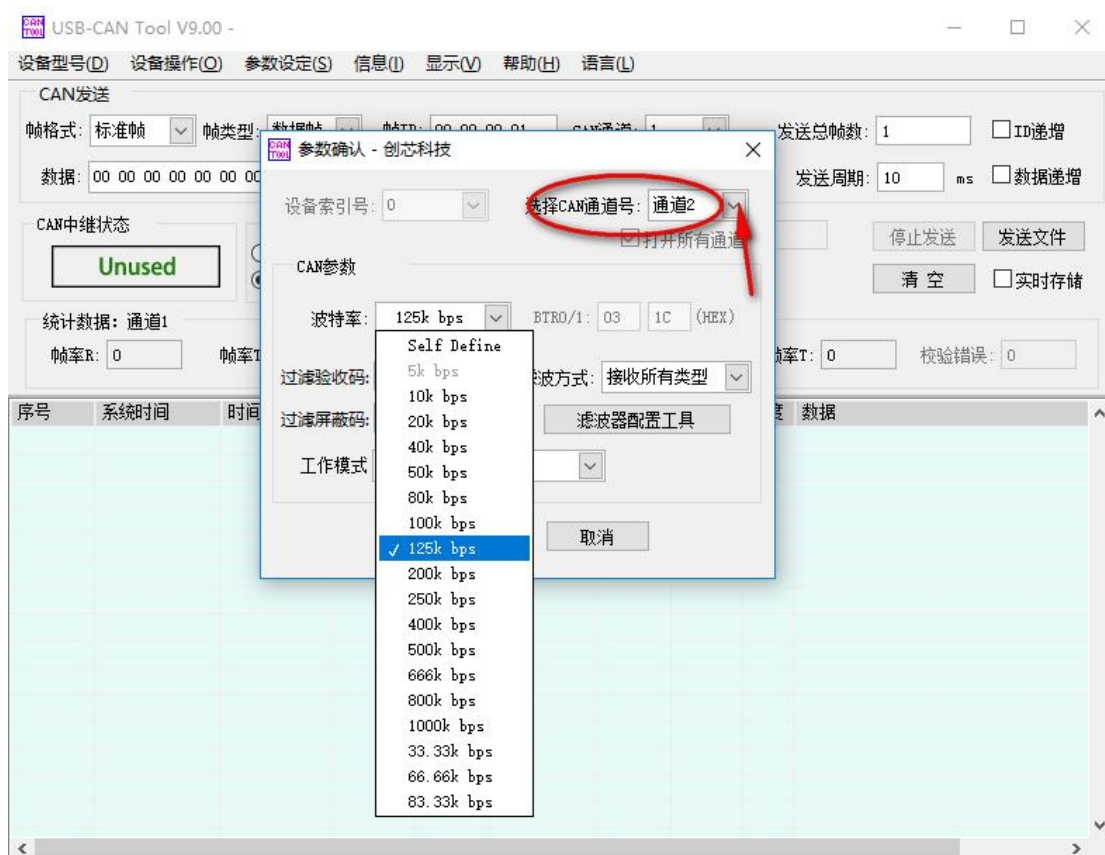
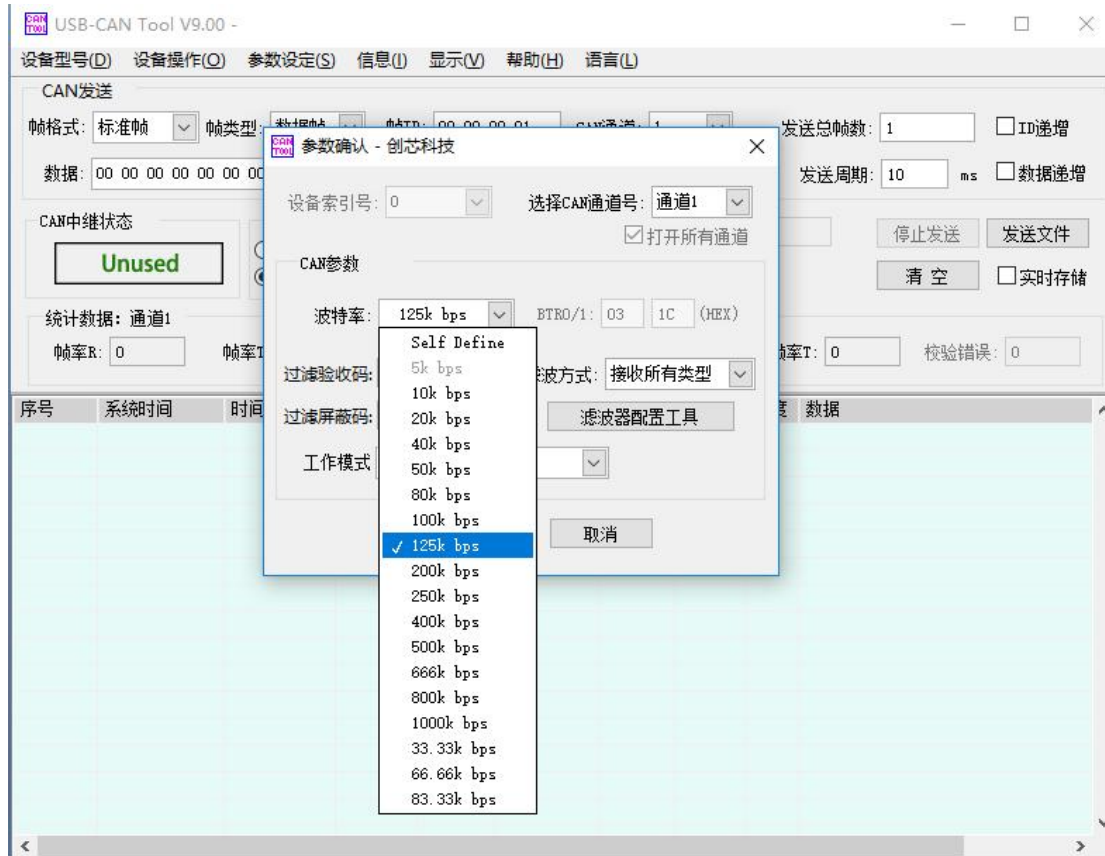
In the initialization parameter dialog box that pops up next, set the parameters of CAN1 channel and CAN2 channel to the same baud rate, and configure the baud rate of CAN1 and CAN2 as 125K, and click "OK":

Note:

1, the baud rate of CAN2 channel needs to switch the channel number to "channel 2" and then configure.

2, USBCAN-2A, USBCAN-2C, CANalyst-II Supreme Edition, Linux Edition, OBD Universal Edition, etc., the baud rate can be configured to 10K-1M, as long as the two channels are the same.

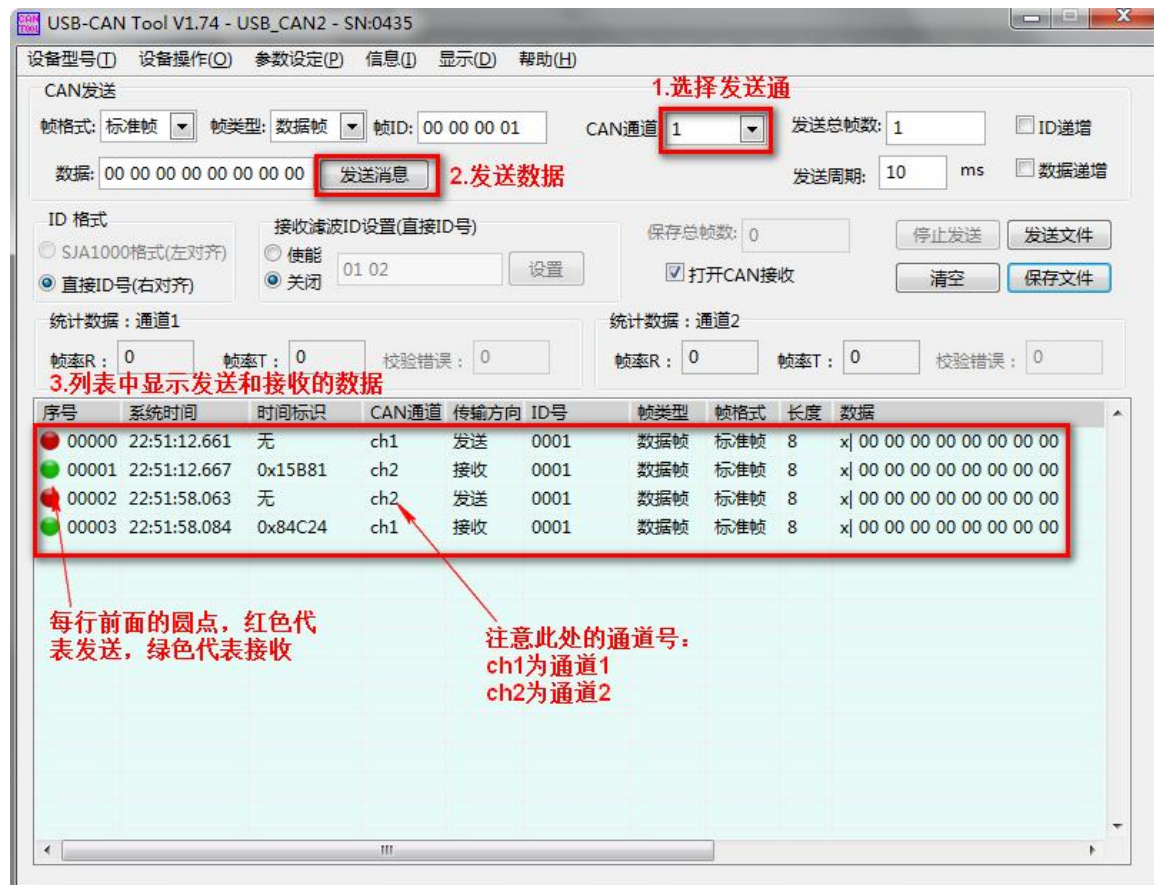
3、CANalyst-II top version with fault-tolerant CAN, the baud rate can only be configured to 10K-125K, because CAN2 channel is fault-tolerant CAN bus baud rate up to 125K.



After opening successfully, select CAN1 channel on the interface, click "Send" button to send a frame of data, at this time, CAN2 channel will receive a frame of the same data, indicating that CAN1 channel sends normal, CAN2 channel receives normal.

Then switch the sending channel to CAN2 channel, click "Send" button to send a frame of data, at this time, CAN1 channel will receive a frame of the same data, indicating that CAN2 channel sends normally and CAN1 channel receives normally.

Note: If no data is received after clicking "Send", please check whether the terminal wiring is firm!



So far, CAN1 channel and CAN2 channel transmission and reception are normal, it means that the USB-CAN bus adapter is normal.

5. When the Test Is Normal by the Above Method, But It Can Not Send and Receive After Accessing the Bus, It Will Be Processed.

After the above steps are tested normally, it means that there is no problem inside the USB-CAN bus adapter, if it still can't receive or send data normally after connecting to the bus, most of them are caused by the physical wiring and baud rate mismatch, please further confirm the following sections:

5.1 Physical Wiring

Strictly follow the above steps for wiring.

Make sure the H and L of both channels are connected reliably.

Make sure the termination resistor is connected correctly.

5.2 Wrong Parameter Matching

Strictly follow the above steps to configure the parameters, if it is used for the first time, use the default values directly. If the configuration has been modified, it is recommended to use the menu "Parameter Setting" -> "Restore Factory Configuration Parameters" and then operate according to the instruction manual.

When the baud rate of the target device is uncertain, you can refer to "*6.plug-in 2: Baud Rate Detection Tool User Manual.pdf*" to detect the bus baud rate.